



RESEARCH ARTICLE

Morphological responses and turf quality expression of Bermudagrass (*Cynodon dactylon* (L.) Pers.) genotypes under salinity stress in simulated sand-based sports field conditions

Akash Elamparuthi¹, Lakshmipathy Muniyappa^{2*}, Rajiv Govindan^{1*}, Md Basit Raza³, D V S Raju¹, Thirumal Subramanian¹, Alluri Sri Sai Mohan², Uma Sowjanya Moturu², Madhavan Srinivasan² & Shakila Sadasivam¹

¹Department of Floriculture and Landscaping Architecture, Sri Ramaswamy Memorial (SRM) College of Agricultural Sciences, Sri Ramaswamy Memorial Institute of Science and Technology, Baburayanpettai, Chengalpattu 603 201, Tamil Nadu, India

²Regional Station, Indian Council of Agricultural Research-Directorate of Floricultural Research, Vemagiri, East Godavari 533 125, Andhra Pradesh, India

³Indian Council of Agricultural Research-Directorate of Floricultural Research, Pune 411 036, Maharashtra, India

*Correspondence email - lakshmipathy.m1@icar.org.in; rajivg@srmist.edu.in

Received: 30 June 2026; Accepted: 31 July 2026; Available online: Version 1.0: 15 September 2026

Cite this article: Akash E, Lakshmipathy M, Rajiv G, Basit RM, Raju DVS, Thirumal S, Alluri SSM, Uma SM, Madhavan S, Shakila S. Morphological responses and turf quality expression of Bermudagrass (*Cynodon dactylon* (L.) Pers.) genotypes under salinity stress in simulated sand-based sports field conditions. Plant Science Today (Early Access). <https://doi.org/10.14719/pst.16436>

Abstract

Salinity stress limits turfgrass performance in urban landscapes, where irrigation is highly dependent on recycled water. This study evaluated the morphological and turf quality responses of six *Cynodon* genotypes, comprising three native accessions (Directorate of Floriculture Research (DFR) RS-2, DFR RS-10, DFR RS-11) and three commercial varieties (Tif Dwarf, Tif Eagle, Selection-1) subjected to four salinity levels (0, 10, 20 and 30 deciSiemens (dS) m⁻¹) in a simulated sand-based sports field profile. Salinity treatments were applied for six weeks after canopy establishment and visual quality rating (VQR), leaf firing, leaf length, clipping yield, dark green colour index (DGCI) and salt secretion were evaluated weekly. Genotype Selection-1 (G6) and DFR RS-11 (G3) consistently maintained acceptable turf quality (VQR ≥ 6), greater leaf elongation, higher canopy greenness and reduced leaf firing under severe salinity stress (30 dS m⁻¹). These responses indicated superior salinity tolerance compared with other genotypes. In contrast, Tif Eagle (G5) exhibited the greatest susceptibility, recording the lowest VQR, DGCI, clipping yield and leaf length, accompanied by the highest salt secretion. These findings identify Selection-1 and DFR RS-11 as promising salinity-tolerant genotypes for sustainable turfgrass cultivation using saline or recycled irrigation water.

Keywords: dark green colour index; genotype screening; salt secretion; salinity tolerance; visual quality rating

Introduction

Bermudagrass (*Cynodon dactylon* (L.) Pers.) is one of the most popular warm season turfgrasses grown worldwide. Its popularity stems from its rapid, dense growth, ability to spread as stolons and underground rhizomes, excellent tolerance to heavy foot traffic and adaptability to warm and hot climates (1, 2). Turfgrasses provide numerous environmental and functional benefits including erosion control, reducing the heat island effect in urban landscapes, enhanced rainwater infiltration, carbon sequestration and improving the aesthetic value of parks and landscapes and the provision of high-quality surfaces for sports and outdoor recreational activities (2, 3).

However, the scarcity of high-quality water resources is becoming increasingly severe due to population growth and expansion of agricultural areas (4). Turfgrass management is water-intensive (5), with bermudagrass requiring approximately 25 to 38 mm of water per week during its growing season, while having a daily water use ranging from 2.5 to 7.5 mm depending on prevailing weather conditions (6, 7). In hot climatic regions, golf courses and

sports fields require more than 1000 mm of water per year, placing considerable pressure on fresh water resources (1). With this high domestic demand, the availability of water for urban green spaces such as gardens, lawns and sports fields is even more limited. Therefore, the recycling and reuse of wastewater for irrigating turfgrass areas in urban landscapes has become an essential strategy for conserving freshwater resources.

Urban areas in India generated about 72368 million litres of sewage per day, of which only 28 % is treated (8). This presents a significant opportunity for wastewater treatment and its reuse in the maintenance of urban green spaces. However, the recycled water contains more dissolved salts, which can adversely affect plant growth when applied as irrigation water. Previous studies have reported that treated wastewater can have high salt concentrations (9). Processes such as reverse osmosis are essential for removing dissolved salts from recycled water (10–12). Despite such treatment, salinity remains one of the major constraints associated with the use of recycled water for turfgrass irrigation. Salinity stress induces a range of morphological changes in Bermudagrasses which includes leaf tip burning (leaf firing), reduced shoot density and leaf length

and decreased biomass accumulation (13). Previous reports suggest that a salt level of 33 deciSiemens (dS m^{-1}) causes a reduction in grass clippings by more than 44 % and an increase in leaf firing by about 38 to 41 % across different grass types (14, 15). However, considerable variation exists among Bermudagrass genotypes in their ability to tolerate saline environments, highlighting the need for systematic screening and selection of salt-tolerant germplasm. Given the increasing scarcity of freshwater resources and the growing reliance on recycled water for urban landscape irrigation, the identification of salt-tolerant Bermudagrass genotypes has become essential for sustainable turfgrass management. Although several Bermudagrass cultivars are widely used on sports fields in India, limited information is available regarding their comparative performance under saline irrigation conditions, particularly in relation to native germplasm resources.

Therefore, this study was to evaluate the salinity tolerance of six *Cynodon* genotypes, including three native accessions and three commercially cultivated varieties commonly used in Indian sports turf systems. The specific objective was to assess their growth, turf quality and physiological responses under different levels of salt stress and to identify genotypes suitable for irrigation using recycled water in water-limited urban environments.

Materials and Methods

Experimental setup and growing conditions

The experiment was conducted at the ICAR-Directorate of Floriculture Research (DFR) Regional Station Andhra Pradesh, India (16.932739°N; 81.788860°E), situated in the East Godavari district within the Eastern Coastal Plains agro-climatic zone. The climate is of the tropical monsoon type, with annual precipitation ranging from 1100 to 1200 mm. The experiment was laid out in a split-plot design with six turfgrass genotypes (G1-*Cynodon* DFR RS-2, G2-*Cynodon* DFR RS-10, G3-*Cynodon* DFR RS-11, G4-Tif Dwarf, G5-Tif Eagle and G6-Selection-1) as subplots and salinity levels (S0-Control (potable water), S1- 10 dS m^{-1} , S2- 20 dS m^{-1} and S3- 30 dS m^{-1}) as main plot with three replications.

Each experimental unit consisted of a polyvinyl chloride (PVC) cylindrical column measuring 100 cm in height and 20 cm in internal diameter. Columns were arranged vertically in an open field, installed 20 cm below ground level to ensure uniform drainage across all replicates. Each column was filled in four distinct layers to simulate a sports field soil profile (Fig. 1). The lower 33.5 cm was filled with sandy loam soil at a bulk density of 1.4 g cm^{-3} , above which a 33.5 cm layer was packed at 1.2 g cm^{-3} of bulk density. Above this, a 3 cm coarse gravel layer was placed and the upper 30 cm was filled with washed river sand amended with vermicompost $360 \text{ g column}^{-1}$.

Planting material and turf establishment

All six turfgrass genotypes were sourced from the turfgrass germplasm block maintained at ICAR-DFR, Regional Station Andhra Pradesh. Stem sprigs bearing two to three intact nodes were cut from actively growing stolons, washed thoroughly to remove any surface contaminants and 20 such sprigs were planted per column at a depth of 1–2 cm in pre-irrigated columns. An irrigation volume was standardised through a preliminary trial. A volume of 750 mL per column per irrigation event was selected as it sustained healthy vegetative growth. Nutrients were added at weekly intervals through water-soluble nitrogen, phosphorus, potassium (NPK) fertiliser (19:19:19) dissolved at 4 g L^{-1} and the salt concentration of the irrigation solution was adjusted to compensate for the increase in electrical conductivity (EC) resulting from fertiliser application, thereby maintaining the target salinity level. The height of the turfgrass was maintained at 2.5 cm (1 inch) by trimming using scissors at weekly intervals.

Salinity stress imposition

Salinity treatments were initiated 60 days after planting (DAP) (once all genotypes achieved full canopy coverage with well-distributed root systems). The EC of the irrigation water was increased gradually in weekly increments of 5 dS m^{-1} , from 5 dS m^{-1} up to each treatment's target EC. Upon reaching the target level, it was maintained continuously for six weeks to allow physiological

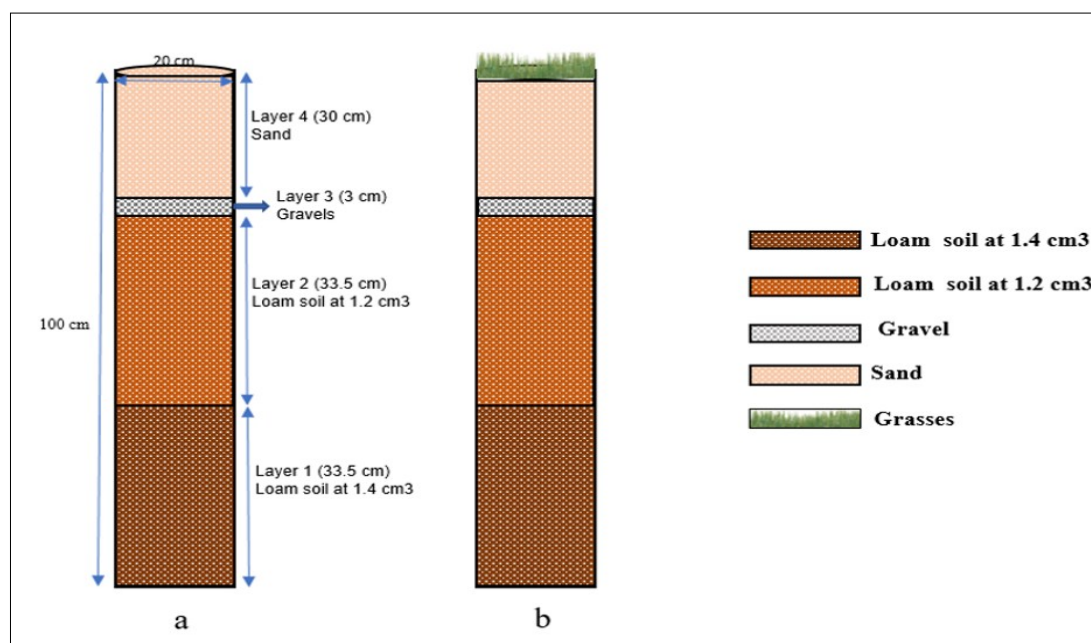


Fig. 1. Simulation of a playing surface with 4 distinct layers. Layer 1-loam soil at 1.4 g cm^{-3} . Layer 2-loam soil at 1.2 g cm^{-3} , Layer 3-gravel, Layer 4-sand layer. Salinity levels: S0-control; S1-10 dS m^{-1} ; S2-20 dS m^{-1} ; S3-30 dS m^{-1} . Bermudagrass Genotypes: G1-*Cynodon* DFR RS-2; G2-*Cynodon* DFR RS-10; G3-*Cynodon* DFR RS-11; G4- Tif Dwarf; G5-Tif Eagle, G6-Selection-1.

acclimatisation while establishing chronic stress. Saline solutions were prepared using a combination of sodium chloride (NaCl), calcium sulphate dihydrate ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) and magnesium chloride (MgCl_2) in a mass ratio of approximately 4:2:2 (g L^{-1}) to simulate treated sewage water and avoid sodicity hazard. All salts were of analytical reagent (AR) grade and solution EC was verified daily using a calibrated digital EC meter ($\pm 0.01 \text{ dS m}^{-1}$).

Observations recorded

Turf quality was assessed by visual scoring on a scale of 1–9, where 1 = fully dead and 9 = fully dark green with excellent density and texture (16). Leaf firing was recorded using a 1–9 visual rating scale, where 1 = severe leaf firing (fully dead) and 9 = no firing (fully green) and healthy leaf, based on leaf colour, texture and burning extent (17). Thus, on this scale a higher leaf-firing score corresponds to less firing (i.e., lower leaf injury and better health), the same direction as the turf quality scale. Leaf length was measured with a ruler from the base to the tip of the third fully matured leaf per shoot. Clippings were collected at each weekly trimming event (cutting height at 1 inch), dried in a hot-air oven at 75°C for 72 hr and the dry weight was recorded in milligrams (18). Dark green colour index (DGCI) was recorded weekly using a Spectrum Field Scout Green Index + reference board and the associated iPhone Operating System (iOS) application. Calibration was performed against the grey colour reference on the board before each measurement. To estimate salt secretion, a marked shoot area was sprayed with distilled water to remove pre-existing salts from the leaf surface prior to measurement. Following a 7-day secretion period, 40 leaves from each column were collected and immediately washed with 10 mL of double-distilled water. The conductivity of the wash solution was determined using a conductivity meter and salt secretion was expressed as $\mu\text{S cm}^{-1} \text{ g}^{-1}$ fresh weight (FW) per 7 days (19).

Statistical analysis

The data were analysed using analysis of variance (ANOVA) and treatment means were separated using the Least Significant Difference (LSD) test at the 5 % significance level ($p \leq 0.05$). Statistical analyses were performed using RAISINS software (version 1.2.0) (19).

Results

Turf quality

The effect of incremental salinity levels in various turf genotypes on visual quality ratings and their interactions, are presented in Table 1. Visual quality rating decreased significantly with increasing salinity levels from S0 (control) to S3 throughout the experimental period. Among the genotypes, G6 consistently recorded the highest VQR. A significant reduction in VQR was observed from the second week onward in all salinity treatments compared with the control. Under moderate salinity (S1) except G5 turf genotype, all the other turfgrass genotypes have maintained minimum acceptable turf quality ($\text{VQR} \geq 6$) during the entire study period of about six weeks, while in G5 the quality has started to decline below 6 VQR from fourth week and scored minimum VQR (5.17) in sixth week. Interestingly G1, G2 and G3 genotypes have a stable VQR from third week onwards. At S2 (20 dS m^{-1}) salinity level G3 and G6 genotypes have maintained minimum VQR (6) in all the weeks, while all other genotypes scored value below 6 VQR, further G5 scored significantly lowest VQR in all the weeks. Under the highest salinity level (S3), G6 maintained the minimum acceptable VQR (≥ 6) throughout the six-week stress period, whereas G5 maintained acceptable quality only up to the fifth week.

Leaf firing

The effects of salinity levels and genotype on leaf firing during six weeks of stress are presented in Table 2. Higher salinity levels induced significantly higher leaf firing in comparison to the control (S0). Genotype G6 consistently recorded the highest leaf-firing rating (i.e., the least leaf injury) throughout the evaluation period. Furthermore, G3 was at par with G6 in first, third, fourth and fifth weeks scoring significantly higher leaf firing ratings. At moderate salinity level (S1) leaf firing was minimum across weeks and none of the grasses scored below 6, but a declining trend of leaf firing score was observed in the genotypes G2 and G5 from first to sixth week. While at higher salinity levels (S2 and S3) the leaf firing score reduced significantly across the genotypes with G2 fell below a rating of 6 from fifth week onwards and G5 started the decline from fourth

Table 1. Effect of varied levels of salinity on turf quality (1–9) of Bermudagrass genotypes (two-way ANOVA, split-plot design; means followed by different superscript letters within a factor differ significantly by critical difference (CD))

	Turf quality Week-1	Turf quality Week-2	Turf quality Week-3	Turf quality Week-4	Turf quality Week-5	Turf quality Week-6
Salinity level						
S0	7.47 ^a	7.47 ^a	7.47 ^a	7.47 ^a	7.47 ^a	7.47 ^a
S1	7.00 ^b	6.92 ^b	6.58 ^b	6.44 ^b	6.39 ^b	6.17 ^b
S2	6.67 ^c	6.42 ^c	6.31 ^c	6.00 ^c	5.86 ^c	5.44 ^c
S3	6.17 ^d	6.14 ^d	5.83 ^d	5.47 ^d	5.19 ^d	4.64 ^d
SE(m)	0.01	0.03	0.02	0.03	0.03	0.02
CD	0.05**	0.12**	0.07**	0.11**	0.10**	0.08**
Grass genotype						
G1	6.83 ^c	6.75 ^d	6.54 ^d	6.33 ^d	6.12 ^c	6.08 ^b
G2	6.62 ^d	6.67 ^e	6.38 ^e	6.12 ^e	6.12 ^c	5.75 ^c
G3	7.00 ^b	6.88 ^c	6.75 ^c	6.75 ^b	6.62 ^b	6.38 ^a
G4	7.25 ^a	7.12 ^a	6.88 ^b	6.46 ^c	6.17 ^c	5.88 ^c
G5	6.25 ^e	6.00 ^f	5.75 ^f	5.54 ^f	5.46 ^d	5.04 ^d
G6	7.00 ^b	7.00 ^b	7.00 ^a	6.88 ^a	6.88 ^a	6.46 ^a
SE(m)	0.02	0.03	0.02	0.04	0.05	0.05
CD	0.05**	0.08**	0.07**	0.12**	0.13**	0.14**
Interaction (S × G)						
SE(m)	0.03	0.05	0.05	0.08	0.09	0.10
CD	0.10**	0.16**	0.14**	0.24**	0.27**	0.27**

Salinity Levels: S0-control; S1-11–10 dS m^{-1} ; S2-22–20 dS m^{-1} ; S3-33–30 dS m^{-1} Bermudagrass genotypes: G1- *Cynodon* DFR RS-2; G2-*Cynodon* DFR RS-10; G3-*Cynodon* DFR RS-11; G4-Tif Dwarf; G5-Tif Eagle; G6-Selection-1. ** indicates significance at 1 % level, NS indicates non-significance. DFR: Directorate of Floriculture Research; SE(m): Standard error of the mean; CD: Critical difference.

Table 2. Effect of varied levels of salinity on Leaf firing (1–9) of Bermudagrass genotypes (two-way ANOVA, split-plot design; a higher score indicates less leaf firing; means followed by different superscript letters within a factor differ significantly by critical difference (CD))

	Leaf firing Week-1	Leaf firing Week-2	Leaf firing Week-3	Leaf firing Week-4	Leaf firing Week-5	Leaf firing Week-6
Salinity level						
S0	8.83 ^a	9.00 ^a	8.94 ^a	9.00 ^a	9.00 ^a	9.00 ^a
S1	8.08 ^b	7.83 ^b	7.50 ^b	7.50 ^b	7.50 ^b	7.50 ^b
S2	7.17 ^c	6.83 ^c	6.75 ^c	6.42 ^c	6.42 ^c	5.81 ^c
S3	6.42 ^d	6.42 ^d	6.17 ^d	5.58 ^d	5.34 ^d	5.08 ^d
SE(m)	0.00	0.00	0.03	0.00	0.00	0.03
CD	0**	0**	0.1**	0**	0.01**	0.1**
Grass genotype						
G1	7.92 ^a	7.62 ^c	7.62 ^b	7.50 ^b	7.50 ^b	7.25 ^b
G2	7.50 ^b	7.50 ^d	7.04 ^c	6.75 ^c	6.75 ^d	6.58 ^d
G3	8.00 ^a	7.75 ^b	7.75 ^a	7.62 ^a	7.63 ^a	7.25 ^b
G4	7.67 ^b	7.75 ^b	7.62 ^b	7.50 ^b	7.38 ^c	7.00 ^c
G5	6.75 ^c	6.62 ^e	6.25 ^d	5.75 ^d	5.50 ^e	5.50 ^e
G6	7.92 ^a	7.88 ^a	7.75 ^a	7.62 ^a	7.62 ^a	7.50 ^a
SE(m)	0.06	0.00	0.03	0.00	0.00	0.03
CD	0.18**	0**	0.1**	0**	0.01**	0.1**
Interaction (S × G)						
SE(m)	0.13	0.00	0.07	0.00	0.01	0.07
CD	0.37**	0**	0.19**	0**	0.02**	0.19**

Salinity Levels: S0-control; S1-11–10 dS m⁻¹; S2-22–20 dS m⁻¹; S3-33–30 dS m⁻¹ Bermudagrass genotypes: G1- *Cynodon* DFR RS-2; G2-*Cynodon* DFR RS-10; G3-*Cynodon* DFR RS-11; G4-Tif Dwarf; G5-Tif Eagle; G6-Selection-1. ** indicates significance at 1 % level, NS indicates non-significance. DFR: Directorate of Floriculture Research; SE(m): Standard error of the mean; CD: Critical difference.

week onwards in S2 level of salinity and most pronounced leaf firing was observed at salinity level S3 in the genotype G5, which showed the leaf firing score less than 6 in all six weeks and the lowest score (2) was recorded in fifth and sixth week. These results indicate considerable genotypic variation in salinity tolerance, with G6 and G3 exhibiting marked resistance to salt-induced leaf injury, whereas G5 was highly susceptible to salinity stress.

Leaf length

Leaf length declined progressively with increasing salinity in all genotypes (Table 3). During week 1 responses varied among genotypes, with values ranging from 2.56 cm in S1 × G2 to 1.05 cm in S3 × G5. Interestingly, G1 and G2 exhibited slightly greater leaf elongation at S1 than under non-saline conditions. By week 2, salinity-induced suppression became more evident, with G5 at S3 showed the lowest value (0.60 cm) in comparison to control G1 at S0 which exhibited highest leaf length (3.09cm). Increasing salinity

progressively reduced leaf elongation across all genotypes, particularly in G3 and G4 under S3 treatment. The reduction intensified during 3rd week when G3 at S0 exhibited the highest elongation (1.86 cm), whereas G5 at S3 recorded the lowest value (0.57 cm). At 4th and 5th week, genotypic differences became more pronounced, with G1 and G6 having relatively greater leaf length under severe salinity, with G6 showing a notable recovery under S3 (1.42 cm in week 5). In contrast, G3, G4 and G5 experienced substantial suppression, with leaf elongation values declining below 0.40 cm under S3. During week 6, two distinct response groups emerged: G1 (1.39 cm) and G6 (1.25 cm) maintained comparatively higher leaf elongation under S3, whereas G2, G3, G4 and G5 exhibited severe reductions, with elongation values below 0.31 cm. These results demonstrate the presence of considerable genotypic variation in leaf growth responses to salinity, with G1 and G6 showing superior adaptation to prolonged salt stress.

Table 3. Effect of varied levels of salinity on Leaf length (cm) of Bermudagrass genotypes (two-way ANOVA, split-plot design; means followed by different superscript letters within a factor differ significantly by critical difference (CD))

	Leaf length Week-1	Leaf length Week-2	Leaf length Week-3	Leaf length Week-4	Leaf length Week-5	Leaf length Week-6
Salinity level						
S0	2.18 ^a	1.96 ^a	1.64 ^a	1.47 ^a	1.53 ^a	1.44 ^a
S1	2.12 ^a	1.73 ^a	1.43 ^b	1.32 ^b	1.30 ^b	1.15 ^b
S2	1.37 ^b	1.30 ^b	1.19 ^c	1.03 ^c	1.06 ^c	0.98 ^c
S3	1.27 ^b	0.90 ^c	0.94 ^d	0.94 ^c	0.70 ^d	0.62 ^d
SE(m)	0.10	0.08	0.01	0.04	0.03	0.03
CD	0.34**	0.27**	0.04**	0.12**	0.12**	0.09**
Grass genotype						
G1	1.86 ^a	1.86 ^a	1.35 ^{ab}	1.39 ^a	1.32 ^a	1.31 ^a
G2	1.98 ^a	1.66 ^{ab}	1.32 ^{ab}	1.35 ^a	1.27 ^a	1.07 ^c
G3	1.81 ^{ab}	1.52 ^b	1.40 ^a	1.22 ^b	1.09 ^b	1.02 ^c
G4	1.58 ^{bc}	1.21 ^c	1.29 ^b	1.13 ^b	0.94 ^c	0.87 ^d
G5	1.47 ^c	1.13 ^c	1.11 ^c	0.92 ^c	0.87 ^c	0.79 ^d
G6	1.72 ^{abc}	1.46 ^b	1.32 ^{ab}	1.15 ^b	1.38 ^a	1.22 ^b
SE(m)	0.09	0.08	0.04	0.04	0.04	0.03
CD	0.26**	0.22**	0.1**	0.12**	0.13**	0.08**
Interaction (S × G)						
SE(m)	0.18	0.15	0.07	0.08	0.09	0.06
CD	NS	0.43**	0.2**	0.24**	0.26**	0.16**

Salinity Levels: S0-control; S1-11–10 dS m⁻¹; S2-22–20 dS m⁻¹; S3-33–30 dS m⁻¹ Bermudagrass Genotypes: G1- *Cynodon* DFR RS-2; G2-*Cynodon* DFR RS-10; G3-*Cynodon* DFR RS-11; G4-Tif Dwarf; G5-Tif Eagle; G6-Selection-1. ** indicates significance at 1 % level; NS indicates non-significance. DFR: Directorate of Floriculture Research; SE(m): Standard error of the mean; CD: Critical difference.

Dark green colour index

The DCGI recorded at fortnightly intervals (2nd, 4th and 6th weeks) showed gradual decline indicating a progressive reduction in turfgrass canopy greenness with increasing salinity (Table 4). The DCGI values of salinity level S0 and S1 were at par in week 2 whereas at week 4 and week 6 all the salinity levels were significantly different from each other, with S3 recording lowest DCGI. These results suggest that prolonged exposure to higher salinity impaired the greenness. The bermudagrass genotype G5 exhibited the lowest DCGI values among all genotypes throughout the study period. At 6th week, G1, G2, G3, G4 and G6 were statistically at par and maintained significantly higher DCGI value. Under S3 during week 6, G5 recorded the lowest DCGI, whereas G6 consistently maintained higher DCGI values throughout the evaluation period.

Biomass production

The rate of biomass accumulation was reflected by clipping dry weights collected at regular intervals to maintain turf height. The total clipping yield data presented in Fig. 2 revealed a significant reduction in biomass productivity with increasing salinity stress. The interaction between salinity level (S) and turfgrass genotype (G) was significant (Fig. 2). Genotype G5 under S3 produced the lowest clipping yield, followed by G5 under S2 and G2 under S2. These treatments exhibited the greatest reduction in biomass accumulation. Among the genotypes, G3 (DFR RS-11) maintained relatively higher clipping yields under both S2 and S3 salinity levels compared with the other genotypes exposed to similar stress conditions, suggesting a greater ability to sustain vegetative growth and biomass production under saline environments. Conversely, G5 (Tif Eagle) exhibited the most pronounced reduction in clipping yield across all salinity treatments, confirming its high sensitivity to salinity stress.

Table 4. Effect of varied levels of salinity on dark green colour index (DCGI) of Bermudagrass genotypes (two-way ANOVA, split-plot design; means followed by different superscript letters within a factor differ significantly by critical difference (CD))

	DCGI Week-2	DCGI Week-4	DCGI Week-6
Salinity level			
S0	0.64 ^a	0.63 ^a	0.67 ^a
S1	0.62 ^a	0.62 ^b	0.63 ^b
S2	0.59 ^b	0.59 ^c	0.60 ^c
S3	0.56 ^c	0.55 ^d	0.52 ^d
SE(m)	0.01	0.00	0.00
CD	0.02**	0.01**	0.01**
Grass genotype			
G1	0.60 ^{abc}	0.62 ^a	0.62 ^a
G2	0.59 ^c	0.59 ^b	0.61 ^a
G3	0.61 ^{ab}	0.62 ^a	0.61 ^a
G4	0.62 ^a	0.59 ^b	0.62 ^a
G5	0.59 ^c	0.57 ^b	0.56 ^b
G6	0.59 ^{bc}	0.59 ^b	0.62 ^a
SE(m)	0.01	0.01	0.01
CD	0.02*	0.02**	0.03**
Interaction (S × G)			
SE(m)	0.02	0.02	0.02
CD	NS	NS	0.06**

Salinity Levels: S0-control; S1-11–10 dS m⁻¹; S2-22–20 dS m⁻¹; S3-33–30 dS m⁻¹ Bermudagrass genotypes: G1- *Cynodon* DFR RS-2; G2-*Cynodon* DFR RS-10; G3-*Cynodon* DFR RS-11; G4-Tif Dwarf; G5-Tif Eagle; G6-Selection-1. * Indicates significance at 5 % level; NS indicates non-significance. DFR: Directorate of Floriculture Research; SE(m): Standard error of the mean; CD: Critical difference.

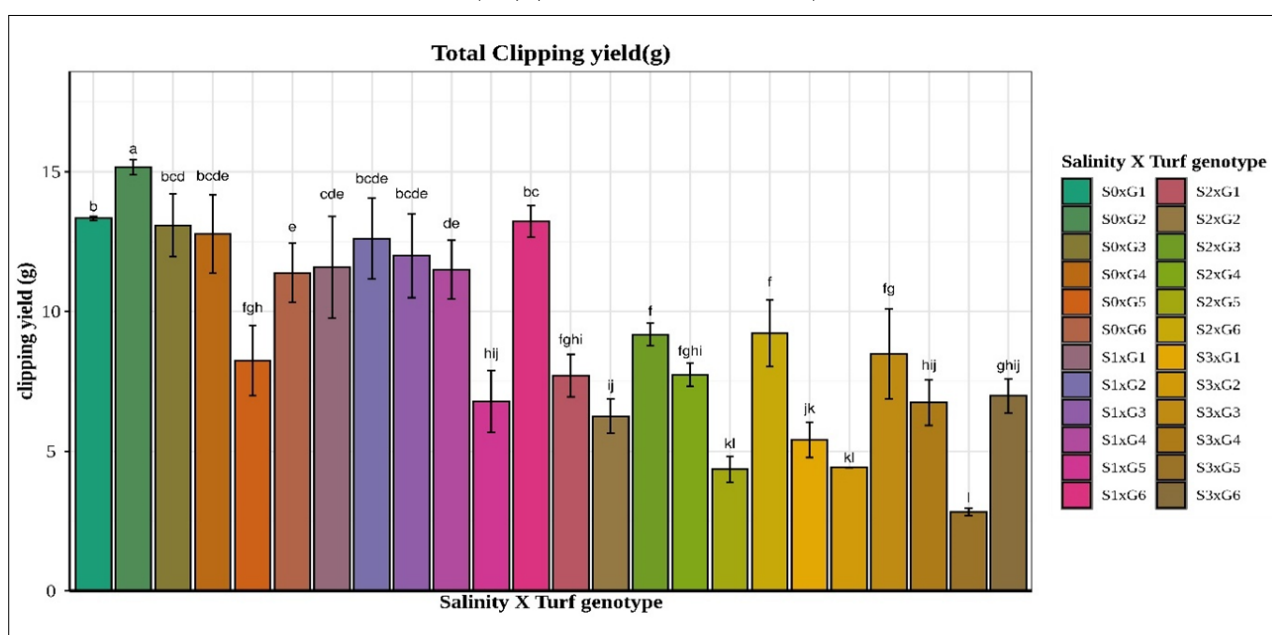


Fig. 2. Effect of salinity levels (S) on different Bermudagrass genotypes (G) on cumulative clipping yield. Data are means from a two-way ANOVA (split-plot design, S × G); error bars/CD as reported in the text denote significance at $P \leq 0.05$. CD: Critical difference. Salinity levels: S0-control; S1-11–10 dS m⁻¹; S2-22–20 dS m⁻¹; S3-33–30 dS m⁻¹. Bermudagrass Genotypes: G1-*Cynodon* DFR RS-2; G2-*Cynodon* DFR RS-10; G3-*Cynodon* DFR RS-11; G4- Tif Dwarf; G5-Tif Eagle; G6-Selection-1.

Salt secretion

Salt secretion increased significantly with increasing salinity (Fig. 3). All the turfgrass genotypes exhibited significantly lower salt secretion at S0 (control) level of salinity in comparison to the salinity treatments. Among the turf genotypes G5 at S3 level of salinity exhibited higher salt secretion than other genotypes at all other levels of salinity, suggesting an enhanced exclusion to cope with excessive salt accumulation. In contrast G1 turfgrass genotype has secreted less salt across all salinity levels, indicating a conservative ionic regulation mechanism.

Discussion

Turf quality and growth parameters

Salinity stress markedly reduced turf quality, confirming its detrimental effects on bermudagrass growth and aesthetic performance. A considerable variation was observed among the bermudagrass genotypes in their response to increasing salinity levels, indicating substantial genetic diversity for salinity tolerance (20). The progressive decline in visual quality ratings (VQR) with increasing salinity concentrations can be attributed to the cumulative physiological and morphological damage induced by salt stress (21). Among the evaluated genotypes, G6 (Selection-1) and G3 (*Cynodon* sp. DFR RS-11) consistently maintained higher turf quality ratings and exhibited lower levels of leaf firing under S2 (20 dS m⁻¹) and S3 (30 dS m⁻¹) salinity treatments. Their ability to sustain visually acceptable turf quality under severe saline conditions suggested the presence of effective tolerance mechanisms that minimise salt-induced injury and help in maintaining the canopy performance. In contrast, G5 (Tif Eagle) showed a rapid decline in VQR with increasing salinity, indicating greater susceptibility to salt stress.

Leaf firing trait is considered one of the most reliable indicators of salinity injury and is frequently used as a primary criterion for assessing turfgrass performance under saline conditions (17). Since the earliest visible symptoms of salt injury appear on the leaves, the extent of leaf firing directly influences turf visual quality (22, 23). The development of leaf firing is largely

attributed to the excessive accumulation of Na⁺ and Cl⁻ ions in leaf tissues, which leads to ion toxicity, chlorophyll degradation, membrane damage and subsequent tissue necrosis (24, 25).

The superior leaf firing scores recorded in G6 and G3 suggest a greater capacity to regulate ionic balance and protect leaf tissues from salinity-induced damage. Conversely, the severe leaf firing observed in G5 suggested inadequate exclusion or compartmentalisation of toxic ions, resulting in accelerated leaf senescence and deterioration of turf quality. Similar findings were reported by previous researchers (26), who observed that Bermudagrass cultivars maintained acceptable turf quality and only moderate leaf firing at salinity levels up to 24 dS m⁻¹. Likewise, some other earlier studies (27) reported significant genotypic variation in turf quality and leaf firing responses among Bermudagrass cultivars under saline conditions, with the cultivar Celebration® maintaining the highest turf quality rating (7.6) at 30 dS m⁻¹ salinity. The present findings corroborate with these reports and further demonstrate the superior salinity tolerance of G6 and G3 compared with the other Bermudagrass genotypes evaluated in this study.

Leaf length and clipping yield declined progressively with increasing salinity levels, indicating the adverse effects of salt stress on vegetative growth and biomass accumulation. Under severe salinity stress (S3- 30 dS m⁻¹), genotypes G1 and G6 maintained comparatively greater leaf elongation throughout the six-week stress period. Similarly, G3 and G6 under S1 (10 dS m⁻¹) and S2 (20 dS m⁻¹) and G3 under S3, recorded higher cumulative clipping yields than the other genotypes. The reduction in leaf elongation and clipping yield reflects a decline in biomass production, as restricted leaf growth reduces the total photosynthetically active leaf area, thereby limiting carbon assimilation and plant productivity (28).

Salinity-induced reductions in leaf growth have been widely reported and are primarily associated with osmotic stress, which decreases cell turgor pressure and restricts cell expansion. Similar findings were reported in *Brassica* (29) where it was noted that salinity adversely affected leaf development and overall plant growth due to osmotic and ionic stress. In turfgrass

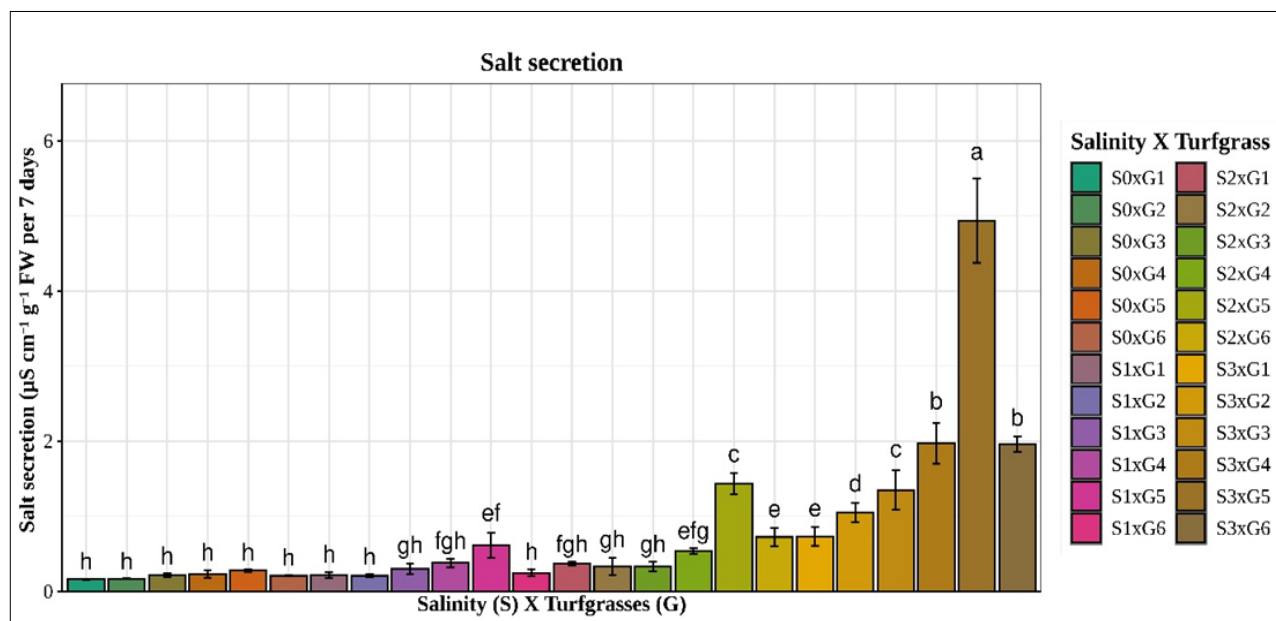


Fig. 3. Effect of salinity levels (S) on salt secretion ($\mu\text{S cm}^{-1} \text{g}^{-1} \text{FW per 7 days}^{-1}$) in different Bermudagrass genotypes (G). Data are means from a two-way ANOVA (split-plot design, S $\times$ G); error bars/CD as reported in the text denote significance at $P \leq 0.05$.

species, previous researchers (30) reported significant reductions in leaf length, leaf area and shoot growth in Zoysia grass and Bermudagrass exposed to saline conditions. Likewise, these studies further (31) observed substantial reductions in clipping yield among Chinese native Bermudagrass accessions under salinity stress, while some other researchers (32) reported significant declines in biomass production and considerable genotypic variation in growth responses among Kentucky bluegrass genotypes subjected to saline environments. The superior leaf elongation maintained by G1 and G6 under severe salinity stress may reflect improved osmotic adjustment of cellular water status, enabling continued growth despite the adverse effects of salinity. According to an earlier report (25), salt-tolerant turfgrass genotypes often maintain growth under saline conditions through improved osmotic regulation, selective ion uptake and efficient water-use mechanisms. Furthermore, the higher clipping yields recorded by G3 and G6 indicate their ability to sustain photosynthetic activity and biomass accumulation under saline conditions, highlighting their superior adaptation and tolerance to salinity stress compared with the more susceptible genotypes like G5.

Physiological characters

Turf quality and DGCI exhibited a strong positive correlation, indicating that canopy greenness is closely associated with overall turf performance (33). Assessment of turfgrass performance under salinity through sensor-based measurements such as DGCI provides a more objective and reliable selection criterion than visual turf quality ratings, which are inherently subjective. In the present study, canopy greenness measured by DGCI declined with increasing salinity, with considerable variation observed among the Bermudagrass genotypes. The higher DGCI values maintained by G6 (Selection-1) under S3 conditions indicate its ability to preserve chlorophyll integrity and sustain photosynthetic activity under salinity stress. In contrast, the consistently lower DGCI values recorded in G5 suggest greater susceptibility to salinity-induced chlorophyll degradation. Similar findings were reported by previous researchers (34), who observed a strong association between turf quality and DGCI in Bermudagrass cultivars exposed to salinity stress. Some other studies (35) similarly reported significant reduction in DGCI under saline conditions and considerable genotypic variation in canopy greenness among *Poa annua* L turf lines.

It is well established that Bermudagrass possesses salt glands on its leaves (36–38), which facilitate ion exclusion through membrane transport proteins (39). Salt secretion in Bermudagrass generally increases with increasing salinity (19, 40), which was evident in the present study. Salt secretion increased by 1.5, 2.7 and 7.7 fold at S1, S2 and S3 respectively, compared with the control (S0). The genotype G5 exhibited the highest salt secretion across all salinity treatments but recorded the lowest VQR, indicating high susceptibility to salt stress. In contrast, earlier studies (13, 27) in Zoysia grass, Bermudagrass and seashore paspalum reported a positive association between salinity tolerance and increased salt secretion, suggesting that greater secretion contributes to enhanced salinity tolerance. In the present study, G6 and G3 also exhibited relatively high salt secretion under S3, second only to G5, while maintaining superior turf quality throughout the study period. This may indicate more efficient Na^+ compartmentalisation in these genotypes, contributing to their enhanced salinity tolerance.

Conclusion

Increasing salinity levels adversely affected turf quality, canopy greenness, leaf growth and biomass production in all the bermudagrass genotypes studied. Among the six genotypes, G6 (Selection-1) and G3 (*Cynodon* sp. DFR RS-11) consistently maintained superior turf quality, canopy greenness and biomass production under high salinity stress, indicating greater salinity tolerance. The results further suggest that salt secretion alone is not a reliable criterion for selecting salt-tolerant genotypes, as neither G3 nor G6 were the highest salt secretors. Further physiological and molecular investigations are warranted to elucidate the mechanisms underlying salinity tolerance, particularly ion compartmentalisation and other adaptive responses, in these promising Indian bermudagrass genotypes.

Acknowledgements

The authors are grateful to the Indian Council of Agricultural Research-Directorate of Floricultural Research, Regional Station, Vemagiri, Andhra Pradesh, for providing the necessary research facilities.

Authors' contributions

AE conducted the research, collected the data and prepared the manuscript. LM was responsible for conceptualisation, resource provision and supervision. RG contributed to conceptualisation. MDBR was involved in conceptualisation, data analysis and preparation of the final draft. DVSR provided plant germplasm and mobilised resources for conducting the study. TS carried out data collection and statistical analysis. ASSM Recourse management and data collection. USM contributed to supervision and data collection. MS performed data analysis and manuscript editing. SS was responsible for conceptualisation and draft editing. All authors read and approved the final manuscript.

Compliance with ethical standards

Conflict of interest: Authors do not have any conflict of interest to declare.

Ethical issues: None

References

1. Harivandi MA, Butler JD, Wu L. Salinity and turfgrass culture. In: Waddington DV, et al., editors. Turfgrass. Agronomy Monograph No. 32. Madison, WI: ASA, CSSA and SSSA; 1992. p. 207–29. <https://doi.org/10.2134/agronmonogr32.c6>
2. Marcum KB. Salinity tolerance of turfgrasses. In: Pessarakli M, editor. Handbook of Plant and Crop Stress. 2nd ed. New York: Marcel Dekker; 2004.
3. Piperno DR, Sues HD. Dinosaurs dined on grass. Science. 2005; 310 (5751):1126–8. <https://doi.org/10.1126/science.1121020>
4. Schiavon M, Baird JH. Evaluation of products to alleviate irrigation salinity stress on bermudagrass turf. Agron J. 2018; 110(5):2136–41. <https://doi.org/10.2134/agronj2018.01.0064>
5. Fu J, Fry J, Huang B. Minimum water requirements of four turfgrasses in the transition zone. HortScience. 2004; 39(7):1740–4. <https://doi.org/10.21273/HORTSCI.39.7.1740>

6. Huang B. Turfgrass water requirements and factors affecting water usage. In: Beard JB, Kenna MP, editors. *Water Quality and Quantity Issues for Turfgrasses in Urban Landscapes*. Ames, IA: Council for Agricultural Science and Technology; 2008. p. 193–205.
7. Wherley B, Heitholt J, Chandra A, Skulkaew P. Supplemental irrigation requirements of zoysiagrass and bermudagrass cultivars. *Crop Sci*. 2014; 54(4):1823–31. <https://doi.org/10.2135/cropsci2013.11.0753>
8. CPCB. Annual Report on Sewage Treatment. Central Pollution Control Board, Ministry of Environment, Forest and Climate Change, Government of India; 2021.
9. Bhardwaj P, Chaturvedi AK, Prasad P. Effect of enhanced lead and cadmium in soil on physiological and biochemical attributes of *Phaseolus vulgaris* L. *Nature and Science*. 2009; 7(8):63–75.
10. Aiello R, Cirelli GL, Consoli S. Effects of reclaimed wastewater irrigation on soil and tomato fruits: a case study in Sicily (Italy). *Agric Water Manag*. 2007; 93(1–2):65–72. <https://doi.org/10.1016/j.agwat.2007.06.008>
11. Rebhun M. Desalination of reclaimed wastewater to prevent salinisation of soils and groundwater. *Desalination*. 2004; 160(2):143–9. [https://doi.org/10.1016/S0011-91-9164\(04\)90004-54-5](https://doi.org/10.1016/S0011-91-9164(04)90004-54-5)
12. Kurup SS, Salem MAM, Cheruth AJ, Sreeramanan S, Purayil FT, Al Amouri AW, et al. Changes in antioxidant enzyme activity in turfgrass cultivars under various saline water irrigation levels to suit landscapes under arid regions. *Commun Soil Sci Plant Anal*. 2017; 48(17):1989–2001. <https://doi.org/10.1080/00103624.2017.1395451>
13. Chen J, Yan J, Qian Y, Jiang Y, Zhang T, Guo H, et al. Growth responses and ion regulation of four warm season turfgrasses to long-term salinity stress. *Sci Hortic*. 2009; 122(4):620–5. <https://doi.org/10.1016/j.scienta.2009.07.004>
14. Dudeck AE, Singh S, Giordano CE, Nell TA, McConnell DB. Effects of sodium chloride on *Cynodon* turfgrasses. *Agron J*. 1983; 75(6):927–30. <https://doi.org/10.2134/agronj1983.00021962007500060017x>
15. Hu L, Huang Z, Liu S, Fu J. Growth response and gene expression in antioxidant-related enzymes in two bermudagrass genotypes differing in salt tolerance. *J Am Soc Hortic Sci*. 2012; 137(3):134–43. <https://doi.org/10.21273/JASHS.137.3.134>
16. Morris KN, Shearman RC. NTEP turfgrass evaluation guidelines. In: NTEP turfgrass evaluation workshop, Beltsville, MD 1998. pp.1–5.
17. Coşkunyel ZB, Selim C, Durak A, Çakır M. Mitigating Salt Stress in Cool-Season Turfgrasses Using Nano Silicon and Humic Acid: A Comparative Evaluation on *Festuca arundinacea* and *Lolium perenne* J Plant Growth Regul. 2026;45(6):4516–30. <https://doi.org/10.1007/s00344-04-026-16-12153-83-8>
18. Qian YL, Bandaranayake W, Parton WJ, Mecham B, Harivandi MA, Mosier AR. Long-term effects of clipping and nitrogen management in turfgrass on soil organic carbon and nitrogen dynamics. *J Environ Qual*. 2003; 32:1694–700. <https://doi.org/10.2134/jeq2003.1694>
19. Marcum KB anderson SJ, Engelke MC. Salt gland ion secretion: a salinity tolerance mechanism among five zoysiagrass species. *Crop Sci*. 1998; 38(3):806–10. <https://doi.org/10.2135/cropsci1998.0011183X003800030031x>
20. Tran TV, Fukai S, Giles HE, Lambrides CJ. Salinity tolerance among a large range of bermudagrasses (*Cynodon* spp.) relative to other halophytic and non-halophytic perennial C4 grasses. *Environ Exp Bot*. 2018; 145:121–9. <https://doi.org/10.1016/j.envexpbot.2017.10.011>
21. Xiang M, Moss JQ, Martin DL, Su K, Dunn BL. Evaluating the salinity tolerance of clonal-type bermudagrass cultivars and an experimental selection. *HortScience*. 2017; 52(1):185–91. <https://doi.org/10.21273/HORTSCI10773-13-16>
22. Marcum KB. Salinity tolerance mechanisms of grasses in the subfamily Chloropidae. *Crop Sci*. 1999; 39(4):1153–60. <https://doi.org/10.2135/cropsci1999.0011183X003900040034x>
23. Lee GJ, Duncan RR, Carrow RN. Salinity tolerance of seashore paspalum ecotypes: shoot growth responses and criteria. *HortScience*. 2004; 39(5):1138–42. <https://doi.org/10.21273/HORTSCI.39.5.1138>
24. Carrow RN, Duncan RR. *Salt-Affected Turfgrass Sites: Assessment and Management*. Chelsea, MI: Ann Arbor Press; 1998.
25. Marcum KB. Use of saline and non-potable water in the turfgrass industry: constraints and developments. *Agric Water Manag*. 2006; 80(1–2):132–46. <https://doi.org/10.1016/j.agwat.2005.07.009>
26. Uddin MK, Juraimi AS, Ismail MR, Othman R, Rahim AA. Growth response of eight tropical turfgrass species to salinity. *Afr J Biotechnol*. 2009; 8(21):5799–806. <https://doi.org/10.5897/AJB09.847>
27. Chavarria M, Wherley B, Thomas J, Chandra A, Raymer P. Salinity tolerance and recovery attributes of warm-season turfgrass cultivars. *HortScience*. 2019; 54(9):1625–31. <https://doi.org/10.21273/HORTSCI13963-13-19>
28. Munns R, Gardner PA, Tonnet ML. Growth and development in NaCl-treated plants. II. Do Na⁺ or Cl⁻ concentrations in dividing or expanding tissues determine growth in barley? *Aust J Plant Physiol*. 1988; 15(4):529–40. <https://doi.org/10.1071/PP9880529>
29. He T, Cramer GR. Absciscic acid concentrations are correlated with leaf area reductions in two salt-stressed rapid-cycling Brassica species. *Plant Soil*. 1996; 179(1):25–33. <https://doi.org/10.1007/BF00011639>
30. Chen Z, Wang ML, Waltz C, Raymer P. Genetic diversity of warm-season turfgrass: seashore paspalum, bermudagrass and zoysiagrass revealed by AFLPs. 2009
31. Chen J, Zong J, Gao Y, Chen Y, Jiang Q, Zheng Y, et al. Genetic variation of salinity tolerance in Chinese natural bermudagrass (*Cynodon dactylon* (L.) Pers.) germplasm resources. *Acta Agric Scand B Soil Plant Sci*. 2014; 64(5):416–24. <https://doi.org/10.1080/09064710.2014.919349>
32. Young JS, Comas LH, Qian Y. Salinity affects root growth of container grown saltgrass. *Int Turfgrass Soc Res J*. 2025; 15:930–9. <https://doi.org/10.1002/its2.70055>
33. Caturegli L, Gaetani M, Volterrani M, Magni S, Minelli A, Baldi A, et al. Normalized difference vegetation index versus dark green colour index to estimate nitrogen status on bermudagrass hybrid and tall fescue. *Int J Remote Sens*. 2019; 40(22):8528–41. <https://doi.org/10.1080/01431161.2019.1641762>
34. Xiang M, Moss JQ, Martin DL, Wu Y. The salinity tolerance of seeded-type common bermudagrass cultivars and experimental selections. *HortTechnology*. 2018; 28(3):276–83. <https://doi.org/10.21273/HORTTECH03975-15-18>
35. Dai J, Schlossberg MJ, Huff DR. Salinity tolerance of 33 greens-type *Poa annua* experimental lines. *Crop Sci*. 2008; 48(3):1187–92. <https://doi.org/10.2135/cropsci2007.06.0320>
36. Marcum KB, Murdoch CL. Salinity tolerance mechanisms of six C4 turf grasses. *J Am Soc Hortic Sci*. 1994; 119(4):779–84. <https://doi.org/10.21273/JASHS.119.4.779>
37. Worku W, Chapman GP. The salt secretion physiology of a chloroid grass, *Cynodon dactylon* (L.) Pers. and its implications. *SINET Ethiop J Sci*. 1998; 21(1):1–16. <https://doi.org/10.4314/sinet.v21i1.18108>
38. Chapman GP, Peat WE. *An Introduction to the Grasses (Including Bamboos and Cereals)*. Wallingford, UK: CAB International; 1992.
39. Zhao C, Zhang H, Song C, Zhu JK, Shabala S. Mechanisms of plant responses and adaptation to soil salinity. *The Innovation*. 2020; 1(1):100017. <https://doi.org/10.1016/j.xinn.2020.100017>
40. Marcum KB, Pessarakli M. Salinity tolerance and salt gland excretion efficiency of bermudagrass turf cultivars. *Crop Sci*. 2006; 46(6):2571–4. <https://doi.org/10.2135/cropsci2006.01.0027>

Additional information

Peer review: Publisher thanks Sectional Editor and the other anonymous reviewers for their contribution to the peer review of this work.

Reprints & permissions information is available at https://horizonpublishing.com/journals/index.php/PST/open_access_policy

Publisher's Note: Horizon e-Publishing Group remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Indexing: Plant Science Today, published by Horizon e-Publishing Group, is covered by Scopus, Web of Science, BIOSIS Previews, Clarivate Analytics, NAAS, UGC Care, etc

See https://horizonpublishing.com/journals/index.php/PST/indexing_abstracting

Copyright: © The Author(s). This is an open-access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution and reproduction in any medium, provided the original author and source are credited (<https://creativecommons.org/licenses/by/4.0/>)

Publisher information: Plant Science Today is published by HORIZON e-Publishing Group with support from Empirion Publishers Private Limited, Thiruvananthapuram, India.